MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 07** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 10: Circles** **MM: 30**

Document Type: Official Step-by-Step Marking Scheme **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

MATHS MANTHAN | (6 × 1 = 6 Marks)

Q1. From a point P, two tangents PA and PB are drawn to a circle with centre O. If $\angle APB = 50°$, then $\angle AOB$ is: [1]

(A) 100° (B) 130° (C) 140° (D) 150°

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 130°

Justification: Since $\angle OAP = \angle OBP = 90°$, in quadrilateral $OAPB$: $\angle AOB = 180° - \angle APB = 180° - 50° = \boxed{130°}$.

Q2. In a right triangle ABC right-angled at C, $BC = 5$ cm and $AC = 12$ cm. The inradius r of the incircle is: [1]

(A) 1.5 cm (B) 2 cm (C) 2.5 cm (D) 3 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 2 cm

Justification: Hypotenuse $AB = \sqrt{5^2 + 12^2} = 13$ cm. Inradius $r = \frac{BC+AC-AB}{2} = \frac{5+12-13}{2} = \boxed{2 \text{ cm}}$.

Q3. If chord AB of length 16 cm of a circle of radius 10 cm is tangent to an inner concentric circle, then the radius of the inner circle is: [1]

(A) 5 cm (B) 6 cm (C) 7 cm (D) 8 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) 6 cm

Justification: Half-chord = 8 cm. Inner radius $r = \sqrt{10^2 - 8^2} = \sqrt{36} = \boxed{6 \text{ cm}}$.

Q4. If tangents PA and PB are drawn to a circle with centre O such that $\angle APB = 120°$, then the relation between OP and tangent AP is: [1]

(A) $OP = AP$ (B) $OP = 2AP$ (C) $OP = \sqrt{3}AP$ (D) $AP = 2OP$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) $OP = 2AP$

Justification: OP bisects $\angle APB \implies \angle OPA = 60°$. In right $\triangle OAP$: $\cos 60° = \frac{AP}{OP} \implies \frac{1}{2} = \frac{AP}{OP} \implies$ **$\boxed{OP = 2AP}$**.

Q5. A line that intersects a circle in two distinct points is called a: [1]

(A) Tangent (B) Secant (C) Radius (D) Normal

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (B) Secant

Justification: By definition, a line intersecting a circle in exactly two distinct points is a **$\boxed{\text{Secant}}$**.

Q6. **Assertion (A):** If two tangents TP and TQ are drawn from an external point T to a circle with centre O, then $\angle PTQ = 2\angle OPQ$.
Reason (R): In $\triangle TPQ$, $TP = TQ$ implies $\angle TPQ = \angle TQP = \frac{1}{2}(180° - \angle PTQ)$, and $\angle OPT = 90°$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: From (R), $\angle OPQ = \angle OPT - \angle TPQ = 90° - (90° - \frac{1}{2}\angle PTQ) = \frac{1}{2}\angle PTQ \implies \angle PTQ = 2\angle OPQ$. Thus both (A) and (R) are true and (R) is the correct explanation of (A).

SECTION B: Very Short Answer Type — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q7. A chord of length 16 cm of a circle of radius 10 cm is tangent to an inner concentric circle. Find the radius of the inner circle. [2]

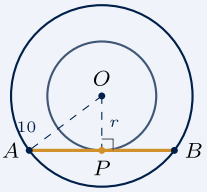
▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let chord $AB = 16$ cm touch the inner circle at P. Since $OP \perp AB$, P bisects AB.

Step 1: Half-Chord Length [0.5 Mark]

$AP = \frac{AB}{2} = \frac{16}{2} = 8$ cm.

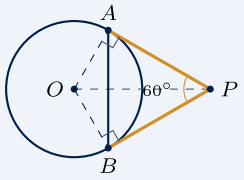
Step 2: Pythagoras Theorem in $\triangle OPA$ [1.5 Marks]



$OP = \sqrt{OA^2 - AP^2} = \sqrt{10^2 - 8^2} = \sqrt{100 - 64} = \sqrt{36} =$ **$\boxed{6\text{ cm}}$**

Q8. If PA and PB are two tangents drawn from an external point P to a circle with centre O such that $\angle APB = 60°$ and $PA = 7$ cm, find the length of chord AB. **[2]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]



Step 1: Equal Tangent Lengths [1 Mark]
By Theorem 10.2, $PA = PB = 7$ cm, so $\triangle PAB$ is isosceles with $\angle PAB = \angle PBA$.

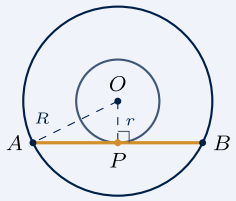
Step 2: Finding Angles & Base Length AB [1 Mark]
$\angle PAB = \angle PBA = \frac{180° - 60°}{2} = 60°$. Since all three angles equal 60°, $\triangle PAB$ is equilateral:

$$\boxed{AB = PA = \mathbf{7}\text{ cm}}$$

SECTION C: Short Answer Type — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q9. Two concentric circles are of radii R and r $(R > r)$. A chord AB of the outer circle touches the inner circle at P. Prove that the length of chord AB is $2\sqrt{R^2 - r^2}$. If $R = 13$ cm and chord length is 24 cm, find r. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



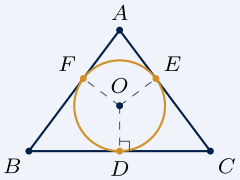
Step 1: Proving Formula for Chord AB [2 Marks]
Radius $OP \perp AB$ (Theorem 10.1), and perpendicular from centre O bisects chord AB $(AP = PB)$. In right $\triangle OPA$, $OA = R, OP = r$:

$$AP = \sqrt{R^2 - r^2} \implies AB = 2AP = \mathbf{2\sqrt{R^2 - r^2}} \quad \text{[Proved]}$$

Step 2: Finding r for $R = 13$ cm, $AB = 24$ cm [1 Mark]
$24 = 2\sqrt{13^2 - r^2} \implies 12 = \sqrt{169 - r^2} \implies r^2 = 169 - 144 = 25 \implies \boxed{r = \mathbf{5}\text{ cm}}$.

Q10. In an isosceles $\triangle ABC$ with $AB = AC = 10$ cm and $BC = 12$ cm, a circle is inscribed. Find the lengths of the tangent segments into which the sides are divided by the points of contact. **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]



Let the incircle touch BC, CA, AB at D, E, F respectively. Let $AF = AE = x$, $BD = BF = y$, $CD = CE = z$.

Step 1: Setting up Side Equations [1.5 Marks]
$x + y = 10, \quad x + z = 10, \quad y + z = 12.$
Subtracting the first two gives $y = z$. Substituting into $y + z = 12 \implies 2y = 12 \implies y = z = 6$ cm.

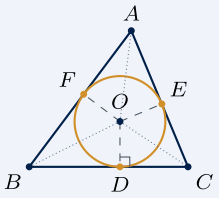
Step 2: Finding x and Segment Lengths [1.5 Marks]
$x = 10 - 6 = 4$ cm. Thus on AB: $AF = 4$ cm, $FB = 6$ cm; on BC: $BD = 6$ cm, $DC = 6$ cm; on AC: $CE = 6$ cm, $EA = 4$ cm.

$$\boxed{\text{Segments: } \mathbf{4}\text{ cm}, \mathbf{6}\text{ cm on } AB, AC; \quad \mathbf{6}\text{ cm}, \mathbf{6}\text{ cm on } BC}$$

Q11. A triangle ABC is drawn to circumscribe a circle of radius 4 cm such that the segments BD and DC into which BC is divided by the point of contact D are of lengths 8 cm and 6 cm respectively. Find the sides AB and AC. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Let the incircle touch AC at E and AB at F. Join OA, OB, OC, OD, OE, OF. Let $AE = AF = x$ cm.



Step 1: Tangent Lengths & Sides [1 Mark]

$BF = BD = 8$ cm, $CE = CD = 6$ cm.

So $a = BC = 14$ cm, $b = AC = x + 6$, $c = AB = x + 8$, and $s = \frac{14+(x+6)+(x+8)}{2} = x + 14$.

Step 2: Area via Heron's vs Inradius [2.5 Marks]

$\text{Area}(\triangle ABC) = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{(x+14)(x)(8)(6)} = \sqrt{48x(x+14)}$.

Also, $\text{Area}(\triangle ABC) = \text{Area}(\triangle OBC) + \text{Area}(\triangle OCA) + \text{Area}(\triangle OAB) = \frac{1}{2}r(a+b+c) = r \cdot s = 4(x+14)$.

Step 3: Solving for x and Sides AB, AC [1.5 Marks]

Equating squares: $48x(x+14) = 16(x+14)^2 \implies 3x = x + 14 \implies 2x = 14 \implies x = 7$ cm (since $x + 14 > 0$).

Therefore, $AB = 7 + 8 = 15$ cm and $AC = 7 + 6 = 13$ cm.

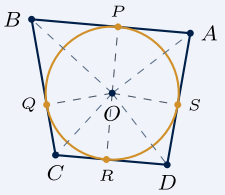
$$\boxed{AB = \mathbf{15}\text{ cm}, \quad AC = \mathbf{13}\text{ cm}}$$

Q12. Prove that the opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle. Using this result, if $\angle AOB = 125°$ where O is the centre, find the measure of $\angle COD$. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: Quad $ABCD$ circumscribes circle (O) touching AB, BC, CD, DA at P, Q, R, S.

To Prove: $\angle AOB + \angle COD = 180°$ and $\angle BOC + \angle AOD = 180°$. [1 Mark]



Proof & Deduction: [3 Marks]

1. Join OA, OB, OC, OD and radii OP, OQ, OR, OS.
2. In $\triangle OAP$ and $\triangle OAS$: $OA = OA$, $OP = OS$, $AP = AS \implies \triangle OAP \cong \triangle OAS$ (SSS) $\implies \angle AOP = \angle AOS$.
3. Similarly, $\angle BOP = \angle BOQ$, $\angle COQ = \angle COR$, $\angle DOR = \angle DOS$.
4. Summing around centre O: $2(\angle AOP + \angle BOP + \angle COR + \angle DOR) = 360° \implies \angle AOB + \angle COD = 180°$. Likewise $\angle BOC + \angle AOD = 180°$. [Proved]

Numerical Application (Given $\angle AOB = 125°$): [1 Mark]

Since opposite sides subtend supplementary angles at O: $\angle AOB + \angle COD = 180° \implies \angle COD = 180° - 125° = \boxed{\mathbf{55°}}$.

Q13. Concentric Circular Running Track and Direct Sprint Chords

A sports stadium features two concentric circular tracks of inner radius $r = 20$ m and outer radius $R = 25$ m sharing a common centre O. A straight sprint corridor AB is painted along a chord of the outer track such that it just grazes (is tangent to) the inner track at point P.

(i) State the angle between radius OP and the sprint chord AB. [1]

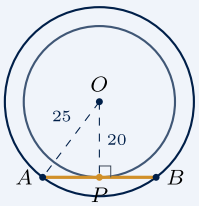
(ii) Find the length AP of the half-chord. [1]

(iii) Calculate the total length of the sprint corridor AB, OR find the area of the track region between the concentric boundaries. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

(i) Angle Between OP and AB: [1 Mark]

Since chord AB touches the inner circle at P, by Theorem 10.1, $OP \perp AB$, so the angle is $\boxed{\mathbf{90^\circ}}$.



(ii) Length of Half-Chord AP: [1 Mark]

In right $\triangle OPA$: $AP = \sqrt{OA^2 - OP^2} = \sqrt{25^2 - 20^2} = \sqrt{225} = \boxed{\mathbf{15}\text{ m}}$.

(iii) Main Choice — Total Length of Sprint Corridor AB: [2 Marks]

Since perpendicular OP bisects chord AB, $AB = 2 \times AP = 2 \times 15 = \boxed{\mathbf{30}\text{ m}}$.

— OR — (Alternative Choice — Area Between Concentric Tracks): [2 Marks]

Area $= \pi(R^2 - r^2) = \pi(25^2 - 20^2) = \pi(225) = \mathbf{225}\pi$ m^2.

$$\boxed{\text{Track Area} = \mathbf{225}\pi \text{ m}^2 \approx \mathbf{706.86} \text{ m}^2 \quad (\text{or } 707.14 \text{ m}^2 \text{ for } \pi = \tfrac{22}{7};\ 706.5 \text{ m}^2 \text{ for } \pi = 3.14)}$$

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